

AYMANE AIT DADS

Data Scientist Intern (PhD-Track) | LLM Fine-Tuning · Ablation Studies · Machine Learning Research

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PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle competition (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE with LoRA ($r=32$, ~888M trainable parameters) and conducting 20+ single-variable ablation studies to isolate causal training factors. Ranked 1st in EURECOM class (0.791 AUC) on an AI-generated image detection task and submitted results to the NTIRE 2026 @ CVPR CodaBench international benchmark, demonstrating research-grade computer vision at academic scale. Core competencies span machine learning, deep learning, NLP, transformer fine-tuning, Python, PyTorch, and data-driven decision-making aligned with financial infrastructure use cases. Available for a Summer 2025 internship in the USA/Canada; eligible to work under OPT/student visa sponsorship.

CORE COMPETENCIES

Machine Learning Research | LLM Fine-Tuning | Ablation Studies | Statistical Modeling | Data-Driven Decision Making | Transformer Architectures | Experimental Design | Quantitative Analysis

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and trained **Random Forest classification and K-Means clustering models** on 100K+ fiber-optic network records to assess quality across thousands of network nodes.
- Built an end-to-end clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation pipelines, and structured ablation of preprocessing steps in Python and Scikit-learn.
- Delivered stakeholder presentations translating **machine learning model outputs** into actionable network maintenance recommendations, bridging technical findings and business decisions.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score of 0.80/1.0** (\$106K+ prize pool) fine-tuning a 30B Mamba-Transformer MoE (3.5B active via MoE routing) using LoRA ($r=32$, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types evaluated by exact match.
- Executed **20+ single-variable ablation studies** isolating completion-only loss (+0.01 AUC vs. full-sequence), sequence packing (-0.16), and LoRA parameter scale (+0.22 over 12-layer baseline), then resolved 5 Blackwell GPU (sm_120) incompatibilities via monkey-patching, sys.modules stubbing, and pure-PyTorch kernel fallbacks.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates ($2e-6$ backbone, $5e-4$ head), and a 3-model ensemble weighted by validation AUC with 10-view TTA.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified through ablation that 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), confirming out-of-distribution generalization as the dominant performance factor across 25+ generator types.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 mixed precision · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Natural Language Processing, Computer Vision, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & NLP: LoRA | QLoRA | SFTTrainer | Hugging Face Transformers | Mamba-Transformer MoE | chain-of-thought | vLLM

Machine Learning & Deep Learning: PyTorch | Scikit-learn | OpenCLIP | TensorFlow | PEFT | TRL | ensemble methods

Data Science & MLOps: Python | SQL | Pandas | NumPy | Power BI | Git | CUDA debugging | FP16/BF16 mixed precision